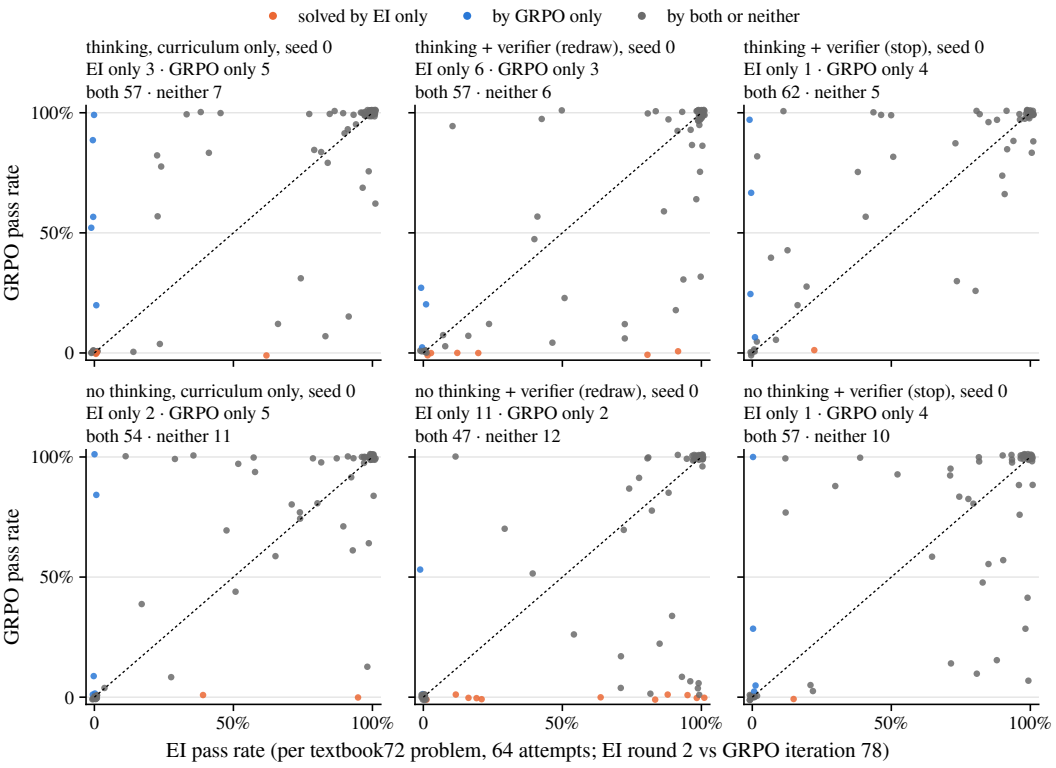
solved by EI only
by GRPO only
by both or neither
thinking, curriculum only, seed 0
EI only 3 · GRPO only 5
both 57 · neither 7
thinking + verifier (redraw), seed 0
EI only 6 · GRPO only 3
both 57 · neither 6
thinking + verifier (stop), seed 0
EI only 1 · GRPO only 4
both 62 · neither 5
no thinking, curriculum only, seed 0
EI only 2 · GRPO only 5
both 54 · neither 11
no thinking + verifier (redraw), seed 0
EI only 11 · GRPO only 2
both 47 · neither 12
no thinking + verifier (stop), seed 0
EI only 1 · GRPO only 4
both 57 · neither 10
GRPO pass rate
100%
50%
0
0
50%
100%
EI pass rate (per textbook72 problem, 64 attempts; EI round 2 vs GRPO iteration 78)